

Problem Solving and Calculations 6

Set 6: Magnetic Fields and Forces

Notes

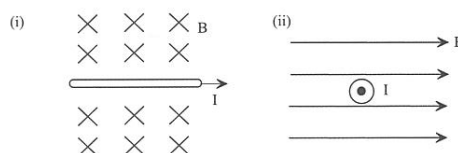
1. Draw the magnetic field distribution for each of the following arrangements:
 - a) bar magnet.
 - b) two parallel (side by side) but separated bar magnets with opposite poles facing each other.
 - c) two parallel but separated bar magnets with like poles facing each other.
 - d) two bar magnets in line (end to end), separated by 50 mm, with opposite poles facing and with a small iron washer placed midway between them.
2. When plotting the magnetic field around a wire carrying a current, a student first ensured that there were no objects made of iron close to the wire. Explain why she did this.
3. A freely suspended current-carrying wire hangs perpendicularly, and parallel to a fixed similar conductor carrying an identical current.
 - a) If both currents flow in the same direction, describe movement, if any, of the suspended wire. Support answer with a diagram.
 - b) What will happen if the currents flow in the opposite direction?
4. The starter motor in a car contains a solenoid (coil) connected to the car's starter switch. Draw a diagram of a solenoid and show the external magnetic field it produces when driver turns the starter switch on. Show the direction of both the current and magnetic field.
5. In a small household electric motor the rectangular coil of wire has 200 turns. The coil is 100 mm long and 30 mm wide. The coil, when it has a current of 0.20 A flowing through it, is perpendicular to a magnetic field of flux density 0.35 T. Find the torque experienced by the coil.
6. Draw the magnetic field pattern and show the relative strengths for:
 - a) current carrying single wire, and
 - b) a current carrying circular coil.
7. The instructions on magnetically-recorded media such as videotapes and computer hard drives state that you should not put them next to electrical power cords. What is the reason behind these instructions?



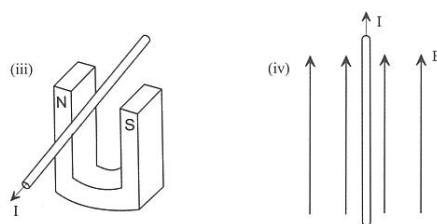
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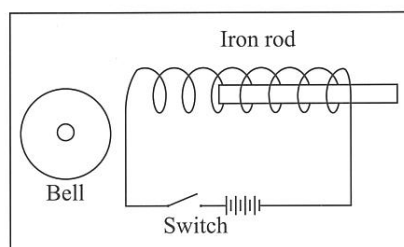
8. For each of the following, indicate the direction of the force, if any, acting on the current carrying wire.



Notes



9. The diagram below represents a circuit for a simple electric door bell. Explain why the bell sounds when you close the switch. Suggest how you can change the arrangement shown in the diagram to make the bell ring louder.



10. An electric motor in a hair dryer draws a current of 10 A when operating at maximum heat and speed. If the side of the armature between the poles of the magnet in the fan motor is 120 mm long, calculate the force acting on the armature when it is perpendicular to the magnetic field of strength 2.0 T.
11. Commercial electric motors may have up to 12 coils or 'windings' with each located in a different position around the armature. Such motors produce a much more even torque than motors with fewer coils. Explain, with the aid of a diagram why this is so.
12. A 75 m long DC transmission cable stretches between two light towers in an east-west direction in a location where the horizontal component of the Earth's magnetic field is 2.5×10^{-5} T. Determine the magnitude and direction of the force that the wire experiences when a 40 A current passes through it.

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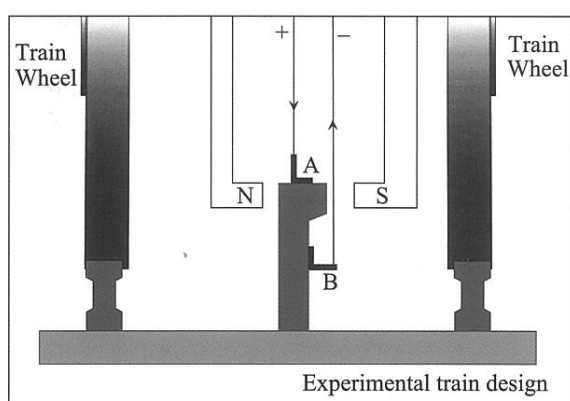
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13. The diagram below shows the design of an experimental electric train.

It has the two poles of a magnet arranged on each side of a steel third rail in the middle of the track. The train has controls that can vary the direction and strength of current that passes between sliding contacts A and B.

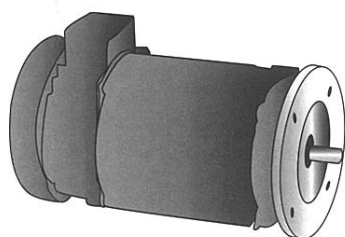
- What is the direction of the magnetic force the magnet exerts on the third rail when current flows from A to B?
- What is the direction of the magnetic force the rail exerts on the magnet?
- In what direction would the train move?

You may find Newton's Third Law of Motion useful in your explanation.



Besides braking, the train could use some of its electric motors as generators to stop the train.

- What are three essential components of an electric motor?
 - How does an electric motor differ from an electric generator?
 - How would the train use its motor to brake?
14. A simple electric motor contains several coils of wire. Draw a diagram of a simple coil and sketch the magnetic field around it that is produced when a current flows through the coil. Show the directions of both the current in the coil and magnetic field produced.



Notes

- 14 a) 383 mm from the balcony
 15 $1.86 \times 10^5 \text{ N}$; $1.95 \times 10^5 \text{ N}$
 20 7.76 kg; 2.57 m from the T_1 end.
 21 60 cm
 23 a) 137 N
 b) lower hinge 172 N toward the door at 36.9° above the horizontal
 upper hinge 172 N toward the wall at 36.9° above the horizontal
 24 a) $1.95 \times 10^3 \text{ N}$
 b) vertical = 109 N upward, horizontal = $1.83 \times 10^3 \text{ N}$ out from wall
 c) $1.84 \times 10^3 \text{ N}$ out from wall at 3.40° above the horizontal
 25 tension = $1.09 \times 10^3 \text{ N}$; force at hinge = 551 N out from the wall and 10.1° below the horizontal
 26 2.07 m from P
 27 a) 156 N
 b) vertical component = 147 N, horizontal component = 156 N
 c) 214 N to right and 43.3° below the horizontal
 28 a) $2.39 \times 10^5 \text{ N}$
 b) $2.97 \times 10^5 \text{ N}$
 29 a) 18.4 kg
 b) 144 N at 26.9° above the horizontal

Electricity and Magnetism

- Set 6 5 0.042 N m
 10 2.4 N
 12 0.075 N downward
 Set 7 1 0.57 V
 2 a) 2.3 A
 5 a) 0.21 mV
 5 b) $2.1 \times 10^{-4} \text{ Wb s}^{-1}$
 7 7.1 mV
 8 a) 39 mV
 8 d) 7.9 mA
 9 a) $8.0 \times 10^{-4} \text{ V}$
 11 225 turns
 12 a) 2.5 m s^{-1}
 12 b) $1.0 \times 10^{-4} \text{ N}$
 13 a) 83 mT
 Set 8 1 a) 60 kW
 1 b) 300 J
 1 c) 150Ω
 2 a) 14.6 m^2
 2 b) 1.25Ω
 3 a) 83 A
 3 b) 18Ω
 6 a) 0.0200 times
 6 b) 10 000 turns
 6 c) 0.0196 times
 7 a) 23Ω
 7 b) $1.3 \times 10^3 \text{ C}$
 8 a) 100 A
 10 a) 31.3 kW
 10 b) 0.31 kW
 12 a) 2.00 kW
 12 b) 0.125 m
 13 a) 6.25 kW
 15 35 km